ESC HEART FAILURE
ESC Heart Failure 2024; **11**: 1795–1801
Published online 12 February 2024 in Wiley Online Library (wileyonlinelibrary.com) **DOI:** 10.1002/ehf2.14719

SHORT COMMUNICATION

Takotsubo syndrome and vaccines: a systematic review

Chenlin Li[1†], Pengyang Li[2†], Bharath Peddibhotla[3], Catherine Teng[4], Ao Shi[5], Xiaojia Lu[6], Peng Cai[7], Qiying Dai[8] and Bin Wang[6,9]*

[1]*Department of Cardiology, Jieyang People's Hospital, Jieyang, Guangdong, China;* [2]*Division of Cardiology, Pauley Heart Center, Virginia Commonwealth University, Richmond, VA, USA;* [3]*VCU School of Medicine, Virginia Commonwealth University, Richmond, VA, USA;* [4]*Division of Cardiology, Department of Medicine, University of Texas San Antonio, San Antonio, TX, USA;* [5]*Faculty of Medicine, St George's University of London, London, UK;* [6]*Department of Cardiology, The First Affiliated Hospital of Shantou University Medical College, Shantou, Guangdong, China;* [7]*Department of Mathematical Sciences, Worcester Polytechnic Institute, Worcester, MA, USA;* [8]*Division of Cardiology, Mayo Clinic, Rochester, MN, USA; and* [9]*Clinical Research Center, The First Affiliated Hospital of Shantou University Medical College, Shantou, Guangdong, China*

Abstract

Aims Takotsubo syndrome (TTS) is a rare complication of vaccination. In this study, we sought to provide insight into the characteristics of reported TTS induced by vaccination.

Methods and results We did a systematic review, searching PubMed, Embase, Web of Science, Ovid MEDLINE, Journals@Ovid, and Scopus databases up to 26 April 2023 to identify case reports or case series of vaccine-induced TTS. We then extracted and summarized the data from these reports. Eighteen reports were identified, with a total of 19 patients with TTS associated with vaccinations. Of the 19 included patients, the majority were female (n = 13, 68.4%) with a mean age of 56.6 ± 21.9 years. Seventeen patients developed TTS after coronavirus disease 2019 vaccination, 14 of whom received an mRNA vaccination. Two cases of TTS occurred after influenza vaccination. Among the 19 patients, 17 (89.5%) completed transthoracic echocardiography and 16 (84.2%) underwent angiography procedures. Seven patients (36.8%) completed cardiac magnetic resonance imaging. The median time to symptom onset was 2 (inter-quartile range, 1–4) days. The most common symptoms were chest pain (68.4%), dyspnoea (57.9%), and digestive symptoms (31.6%). A total of 57.9% of patients developed nonspecific symptoms such as fatigue, myalgia, diaphoresis, and fever. Among the 16 reported cases of TTS, 15 patients (93.8%) exhibited elevated cardiac troponin levels, while among the nine reported cases, eight patients (88.9%) had elevated natriuretic peptide levels. All patients had electrocardiographic changes: ST-segment change (47.1%), T-wave inversion (58.8%), and prolonged corrected QT interval (35.3%). The most common TTS type was apical ballooning (88.2%). Treatment during hospitalization typically included beta-blockers (44.4%), angiotensin-converting enzyme inhibitors/angiotensin receptor blockers (33.3%), and diuretics (22.2%). After treatment, 81.3% of patients were discharged with improved symptoms. Among this group, nine patients (56.3%) were reported to have recovered ventricular wall motion during follow-up. Two patients (12.5%) died following vaccination without resuscitation attempts.

Conclusions TTS is a rare but potentially life-threatening complication of vaccination. Typical TTS symptoms such as chest pain and dyspnoea should be considered alarming symptoms, though nonspecific symptoms are common. The risks of such rare adverse events should be balanced against the risks of infection.

Keywords Takotsubo syndrome; COVID-19 vaccine; Influenza vaccine; Systematic review

Received: 4 January 2023; Revised: 29 December 2023; Accepted: 29 January 2024
**Correspondence to: Bin Wang, Department of Cardiology, The First Affiliated Hospital of Shantou University Medical College, 57 Changping Road, Shantou, Guangdong 515041, China. Tel: +86-075488905399; Fax: 075488259850. Email: wangbin_pku@126.com*
[†]*These authors contributed equally to the writing of this article and share primary authorship.*

Introduction

With the coronavirus disease 2019 (COVID-19) pandemic, there is growing interest in the complications of mRNA vaccination. While vaccines, regardless of the underlying mechanism, are considered safe and highly effective in infectious disease prevention,[1] their cardiac side effects have caught the public eye. An increasing number of studies have reported takotsubo syndrome (TTS) development in the setting of vaccination.[2–19] TTS triggered by vaccines is extremely rare, and published literature available so far is limited to case reports. There is no large-scale systematic review examining the association between vaccines and TTS.

Aims

In this study, we sought to summarize the characteristics of TTS triggered by vaccination by literature review.

Methods

Search strategy and selection criteria

We searched PubMed, Embase, Web of Science, Ovid MEDLINE, Journals@Ovid, and Scopus databases up until 26 April 2023 to identify publications that reported the development of TTS after vaccination. The search strategy consisted of a combination of the following keywords: vaccine OR vaccination AND apical ballooning syndrome OR broken heart syndrome OR stress cardiomyopathy OR takotsubo cardiomyopath* OR takotsubo syndrome OR takotsubo cardiomyopathy OR takotsubo OR ampulla cardiomyopathy OR neurogenic pulmonary edema OR stress induced cardiomyopathy. Case reports and series studying TTS after vaccination were included in the initial screening. We conducted a thorough review of the full text of the remaining articles and reassessed the reliability of the TTS diagnosis by considering patient clinical characteristics, laboratory findings, and cardiac test results. The screening processes were performed on EndNote X9 software (Clarivate Analytics).

Eligibility criteria

All enrolled studies included at least one case of vaccine-induced TTS, regardless of the type of vaccine and dose. Enrolled studies were limited to case reports and case series. Randomized controlled trials, narrative and systematic reviews, and studies with insufficient data were excluded.

Study selection

Three hundred twenty-six publications were identified after the systematic search, with 187 being duplicates. One hundred fourteen studies were further excluded after screening for title and abstract. Of the remaining 25 articles, four studies focused on complications other than vaccination,[20–23] one study focused on non-TTS cardiac complications after vaccination,[24] and two studies were not case reports or case series.[25,26] Those studies were excluded as a result. Finally, 18 studies met the inclusion criteria and were included in this review (*Figure 1*).

Data extraction and quality assessment

Authors CL and PL extracted data using a predefined template and performed the quality assessment from the selected studies. The extracted information included the case summary, patient demographics, initial presentations, cardiac laboratory, imaging findings, and clinical outcomes. Categorical variables were expressed as counts and percentages, and continuous variables were expressed as mean ± standard deviation. Time from vaccination to symptom onset was expressed as median and inter-quartile range. The inconsistencies were resolved by the third author (XL). Studies were scored as 'yes', 'no', or 'not applicable', following the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for quality assessment of case reports based on an eight-item scale. The quality score was graded as poor, fair, and good if the quality score was 0–2, 3–5, and 6–8.

Results

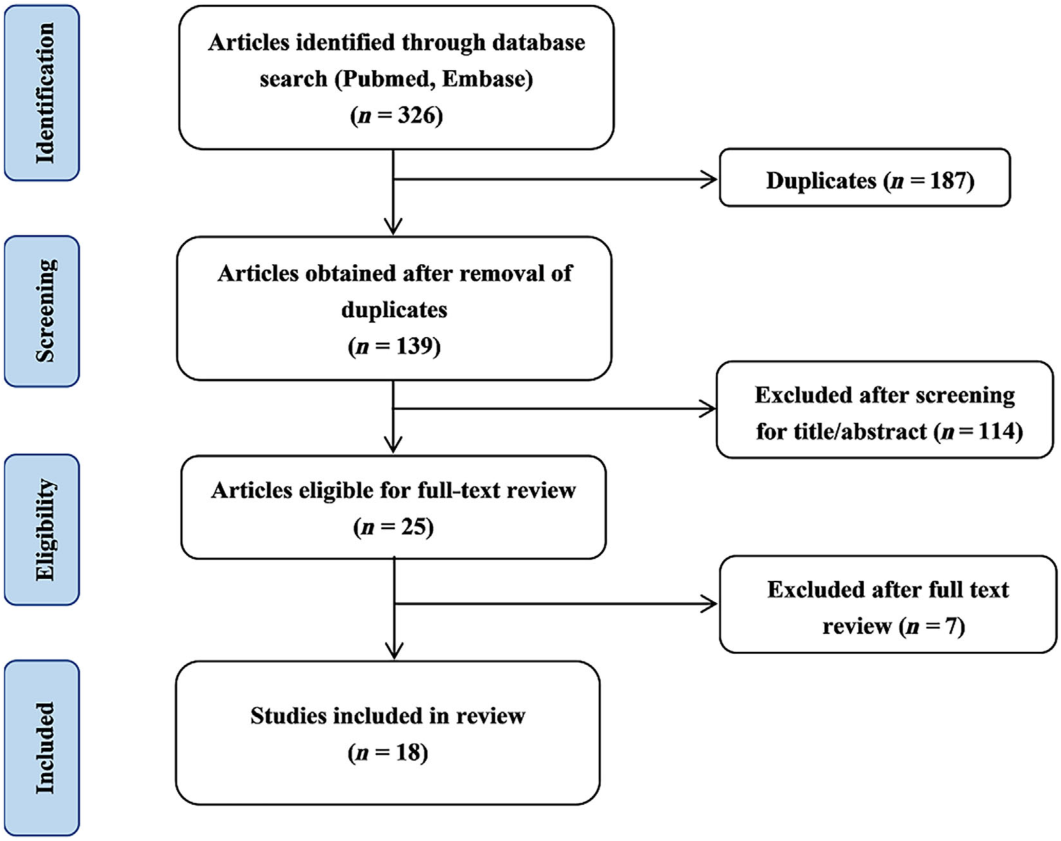
Clinical characteristics

A total of 19 TTS patients associated with vaccines were found. The majority of the patients were female (n = 13, 68.4%) with a mean age of 56.6 ± 21.9 years. The most common comorbidity was hypertension (n = 4, 26.7%). Seventeen patients developed TTS after COVID-19 vaccination, and 14 of them received the mRNA vaccine. The other two patients developed TTS within 3 days after influenza vaccination. The median time from vaccination to symptom onset was 2 (inter-quartile range, 1–4) days. Common presentations included chest pain (68.4%), dyspnoea (57.9%), and digestive symptoms (31.6%). More than half of the patients had other nonspecific symptoms (57.9%), such as fatigue, myalgia, diaphoresis, and fever. Details of the characteristics of vaccine-induced TTS patients are presented in Supporting Information, *Table S1*.

Cardiac laboratory and imaging findings

Most of the patients included in the studies underwent a comprehensive series of cardiac laboratory and imaging tests as part of the diagnostic process for TTS. Out of the 19 patients reported in the included studies, two teenage patients (10.5%) died following vaccination. Although no other cardiac test findings were present, the diagnosis of TTS was confirmed through histopathological cardiac findings observed

Figure 1 Flow chart of the study selection.



during autopsy. The remaining 17 patients (89.5%) have all completed the transthoracic echocardiography (TTE) test, among whom 15 patients (88.2%) presented with typical TTS changes of hypokinesia of the apical walls. In terms of cardiac troponin and natriuretic peptide levels, among the reported cases, 16 patients had their cardiac troponin levels assessed, and positive findings were observed in 15 patients (93.8%). Additionally, nine patients had their natriuretic peptide levels evaluated (including brain natriuretic peptide or N-terminal prohormone brain natriuretic peptide), with positive findings in eight patients (88.9%) (Supporting Information, *Table S2*). Out of the total number of patients, 16 individuals (84.2%) underwent angiography procedures, including coronary catheterization angiography (15 patients, 78.9%) and coronary computed tomography angiography (2 patients, 10.53%), with one patient undergoing both tests. Notably, none of these patients exhibited evidence of coronary artery occlusion or acute plaque rupture. The remaining patient did not undergo coronary angiography or computed coronary angiography because he was unsuitable for intravenous sedation, but the possibility of coronary artery disease can also be basically ruled out because of the identity of the adolescent boy and the self-limiting course of the disease. Seven patients (36.8%) have completed cardiac magnetic resonance (CMR) to further clarify the diagnosis of TTS, among whom four patients (57.1%) showed no late gadolinium enhancement (LGE) evidence and three patients (42.9%) showed non-ischaemic LGE pattern. The details of the findings can be found in *Table 1* and Supporting Information, *Table S3*.

Treatments and outcomes

During hospitalization, the common treatments for TTS patients included beta-blockers (44.4%), angiotensin-converting enzyme inhibitors (ACEIs)/angiotensin receptor blockers (ARBs) (33.3%), and diuretics (22.2%). Acute coronary syndrome (ACS) protocol of dual antiplatelet therapy was generally the initial management for TTS patients because of their similar clinical course, but it was not emphasized after excluding coronary artery disease. Two teenage boys, as reported by Gill *et al*., were found dead at home after the second dose of COVID-19 vaccine without attempted resuscitation. Additionally, one patient, as reported by Ceresa *et al*., underwent Dor's procedure to repair the left ventricular aneurysm.

Table 1 Summary of clinical characteristics and outcomes in vaccine-induced TTS patients

Characteristics	
Number of patients	19
Age (years)	56.6 ± 21.9
Sex	
Male	6/19 (31.6%)
Female	13/19 (68.4%)
Quality of included studies	
Poor	0
Fair	6/18 (33.3%)
Good	12/18 (66.7%)
Type of vaccine	
COVID-19 vaccine	17/19 (89.5%)
mRNA vaccine	14/17 (82.3%)
Adenovirus vector vaccine	2/17 (11.8%)
Unknown	1/17 (5.9%)
Influenza vaccine	2/19 (10.5%)
Dose of vaccine	
1st	7/16 (43.2%)
2nd	8/16 (50.0%)
3rd	1/16 (6.3%)
Time from vaccination to symptom onset (days), median (IQR)	2 (1–4)
Symptom	
Chest pain	13/19 (68.4%)
Dyspnoea	11/19 (57.9%)
Digestive symptoms	6/19 (31.6%)
Others (fatigue/myalgia/diaphoresis/fever)	11/19 (57.9%)
Laboratory test	
Elevated CK or CK-MB	3/7 (42.9%)
Elevated cardiac troponin	15/16 (93.8%)
Elevated BNP/NT-proBNP	8/9 (88.9%)
Cardiac test	
ECG	17/19 (89.5%)
TTE	17/19 (89.5%)
Coronary angiography	15/19 (78.9%)
Coronary computed tomography angiography	2/19 (10.5%)
Ventriculography	7/19 (36.8%)
CMR	7/19 (36.8%)
Autopsy	2/19 (10.5%)
ECG	
ST-segment change	8/17 (47.1%)
ST-segment elevation	4/17 (23.5%)
ST-segment depression	3/17 (17.6%)
Unspecified	2/17 (11.8%)
T-wave inversion	10/17 (58.8%)
Prolonged QTc interval	6/17 (35.3%)
Poor anterior R-wave progression	1/17 (5.9%)
Sinus tachycardia	1/17 (5.9%)
RBBB	2/17 (11.8%)
Ventriculography/TTE	
LVEF $<$ 40%	8/17 (47.1%)
Type of TTS	
Apical	15/17 (88.2%)
Midventricular	1/17 (5.9%)
Basal	1/17 (5.9%)
Focal	0
Biventricular	0
Coronary artery occlusion or acute plaque rupture	
Yes	0/16
No	16/16 (100%)
CMR	
No LGE evident	4/7 (57.1%)
Non-ischaemic LGE pattern	3/7 (42.9%)
Myocardial oedema	4/7 (57.1%)
Treatment	
ACEI/ARB	3/9 (33.3%)
Beta-blockers	4/9 (44.4%)
Diuretics	2/9 (22.2%)
Antiplatelet therapy	0
Outcome	
Improved symptom	13/16 (81.3%)
Recovered LV wall motion	9/16 (56.3%)
Left ventricular aneurysm	1/16 (6.3%)
Death	2/16 (12.5%)

ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BNP, brain natriuretic peptide; CK, creatine kinase; CK-MB, creatine kinase myocardial band; CMR, cardiac magnetic resonance; COVID-19, coronavirus disease 2019; ECG, electrocardiogram; IQR, inter-quartile range; LGE, late gadolinium enhancement; LV, left ventricular; LVEF, left ventricular ejection fraction; NT-proBNP, N-terminal prohormone brain natriuretic peptide; QTc, corrected QT; RBBB, right bundle branch block; TTE, transthoracic echocardiography; TTS, takotsubo syndrome.

Among the remaining 13 patients (81.3%), they were discharged with improved symptoms, and during follow-up, 9 patients (56.3%) were reported to have recovered ventricular wall motion (Supporting Information, *Table S4*).

Discussion

In this study, we summarized the characteristics of TTS triggered by vaccination by literature review. We found that TTS followed by vaccination is extremely rare with only 19 cases published thus far, and most of the cases were reported after receiving COVID-19 mRNA vaccine. The majority of patients restored ventricular wall motion and cardiac systolic function after treatment.

Of the 19 patients from the included studies, 13[2–9,11,12,14,16,17] of them adhere to the established diagnostic criteria of TTS according to the diagnostic algorithm of TTS published by the European Heart Journal in 2018,[27] but the remaining may need further cardiac tests to confirm the diagnosis. Three patients presented with typical clinical symptoms of TTS and had typical apical ballooning of the left ventricle by echocardiography or ventriculography, but lacking CMR to exclude acute myocarditis given the presence of corresponding red flags such as fever[15,19] or elevated c-reactive protein.[13] Another patient presented with hypokinesia of the inferior wall after vaccination and subsequently developed a left ventricular aneurysm. Ceresa *et al*. believed that the atypical TTS related to COVID-19 vaccination was the most likely hypothesis.[18] However, pericardial effusion detected by TTE and a lack of CMR did not rule out the left ventricular aneurysm as a very rare complication of myocardial infarction with nonobstructive coronary artery disease. Gill *et al*. reported on two teenage boys with sudden death after vaccination, and their histopathological results were consistent with cardiac pathological changes mediated by epinephrine in TTS

DOI: 10.1002/ehf2.14719

patients.[10] However, hypersensitivity myocarditis was another possible diagnosis.[28] Hypersensitivity reaction was also put in the differential diagnosis by Gill *et al*., though they suggested that infrequency or lack of eosinophils in perivascular and interstitial inflammation would be unusual. Considering that these 'red flags' may also manifest in patients following vaccination, and given the relatively small number of reported cases of vaccine-induced TTS, we have included these instances in our study to provide a comprehensive characterization of vaccine-induced TTS.

Minciullo *et al*. reported a patient diagnosed with TTS and Kounis syndrome overlap, who developed lower and lateral akinesia with no evidence of coronary artery obstruction after the initial allergic reaction after vaccination and the consecutive epinephrine administration. It is difficult to clearly define the trigger of TTS in this patient. The mRNA vaccine may be involved in the onset of TTS, but physical stress and epinephrine drugs may also play a role.[17]

The fact that most of the reported TTS-vaccine cases were triggered by COVID-19 vaccines should be interpreted with caution. Due to the overwhelming public debate and interest in complications after COVID-19 vaccines, the cases may be subject to report bias. Separately, despite public education, there is significant anticipatory stress prior to this particular vaccination administration due to its unknown and perceived side effects. Though at least one-third of TTS cases were triggered by emotional stressors,[29,30] it is unclear that anticipatory stress could have triggered the TTS in the 17 COVID-19 vaccine cases.

mRNA COVID-19 vaccines have been implicated in the majority of TTS cases, possibly due to the reactivity of mRNA vaccines and resulting inflammation.[26,31] mRNA vaccine consists of lipid nanoparticle shell and mRNA molecules that encode protein antigens. Research has demonstrated that mRNA vaccines can elicit a robust innate immune response, particularly during secondary immunization.[32] Furthermore, upon administration into muscle tissue, the mRNA is translated into the target protein, such as the spike protein of SARS-CoV-2, within the cytoplasm, stimulating both cellular and humoral immune responses.[33] However, despite additional modifications and encapsulation of mRNA vaccines in lipid nanoparticles during the production process, it is challenging to completely eliminate undesired immune responses due to the inherent instability of mRNA molecules under physiological conditions.[34] Minor variations in biophysical characteristics, such as particle size and homogeneity, can impact physicochemical stability and immunogenicity.[35] Consequently, an escalated inflammation cascade and increased levels of inflammatory factors, such as tumour necrosis factor-α (TNF-α) and interleukin-6 (IL-6), are commonly observed following vaccination.[36] Systemic inflammation can potentially contribute to TTS development by promoting vascular endothelial dysfunction, which is one of the proposed mechanisms of TTS.[37,38]

Our literature review suggests that when cardiac symptoms such as chest pain and dyspnoea arise after vaccination, one should take them seriously as they could potentially be an alarming symptom of TTS. This is in addition to the commonly known side effects of vaccines (COVID-19 vaccine in particular), such as fatigue, diaphoresis, and digestive symptoms. On the other hand, the absence of cardiac symptoms does not mean absolute safety. Two teenage boys died after vaccination without typical cardiac symptoms such as chest pain, and they were diagnosed with TTS after autopsy. TTS, once regarded as a benign condition with a self-limiting clinical course, has been increasingly recognized for its high risk of acute complications comparable with those seen in patients with ACS. In the International Takotsubo Registry study, the risk of a composite of catecholamine use, cardiogenic shock, the use of invasive or noninvasive ventilation, cardiopulmonary resuscitation, and death was 19.1% in TTS patients vs. 19.3% in a control group of patients with ACS.[39] A National Inpatient Sample study of 88 849 TTS patients found that frequency of complications, including acute heart failure and shock, acute kidney injury, arrhythmia, vascular access complications, ventricular and papillary muscle rupture, tamponade, and stroke, was higher (38.2% vs. 32.6%) compared with acute myocardial infarction.[40] These studies collectively highlight the necessity for a more cautious approach in managing TTS. Furthermore, the deaths of two teenage boys in this study, attributed to TTS following vaccination, further reinforce this point. The potential adverse event required us to pay more attention to such cases. The detail of the prognosis of TTS triggered by vaccination needs to be explored with more data in further studies.

In comparison, there have been more reported cases of TTS associated with COVID-19 than TTS cases related to vaccines. José *et al*. conducted a literature review and identified 71 subjects with TTS and COVID-19 across 35 published reports, in which 18.3% of patients died.[41] Another systematic analysis, which included 52 TTS cases from a total of 40 articles, reported an all-cause mortality rate of 36.5% in TTS patients with COVID-19.[42] Additionally, the European Association of Cardiovascular Imaging found that 2% of 1216 COVID-19 patients included in their analysis had concomitant diagnoses of TTS based on echocardiography findings.[43] Notably, there is currently no study specifically investigating the incidence of TTS following COVID-19 vaccination. Therefore, it is not possible to directly compare the incidence risk of TTS between COVID-19 infection and COVID-19 vaccination. However, it is crucial to consider that COVID-19 can cause significant damage to the respiratory system and lead to various cardiac complications, including myocarditis and pericarditis.[44] Given these factors, it is essential to emphasize that despite the risk of TTS and associated complications, COVID-19 vaccination is essential and highly beneficial for the population in preventing the widespread transmission of the virus.

Conclusions

This review highlights that TTS is a rare but potentially life-threatening complication of vaccines. When TTS occurs after vaccination, the presentations are often typical, with elevated troponin and electrocardiogram (ECG) changes. In most cases, cardiac function can be restored, but some TTS cases can be fatal. Long-term outcomes and future cardiovascular risks of such patients should be explored in future studies.

Conflict of interest

None declared.

Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1. Characteristics and outcome of vaccines-induced TTS patients.
Table S2. Laboratory findings of vaccines-induced TTS patients.
Table S3. Cardiac test finding of vaccines-induced TTS patients.
Table S4. Supporting Information.
Table S5. JBI Critical Appraisal Checklist for Case Reports.

References

1. Conklin L, Hviid A, Orenstein WA, Pollard AJ, Wharton M, Zuber P. Vaccine safety issues at the turn of the 21st century. *BMJ Glob Health* 2021;**6**:e004898. doi:10.1136/bmjgh-2020-004898
2. Toida R, Uezono S, Komatsu H, Toida T, Imamura A, Fujimoto S, *et al*. Takotsubo cardiomyopathy after vaccination for coronavirus disease 2019 in a patient on maintenance hemodialysis. *CEN Case Report* 2022;**11**:220-224. doi:10.1007/s13730-021-00657-z
3. Stewart C, Gamble DT, Dawson D. Novel case of takotsubo cardiomyopathy following COVID-19 vaccination. *BMJ Case Rep* 2022;**15**:e247291. doi:10.1136/bcr-2021-247291
4. Yamaura H, Ishikawa H, Otsuka K, Kasayuki N. Reverse takotsubo cardiomyopathy as a cause of acute chest pain in a young woman following COVID-19 vaccination. *Circ Cardiovasc Imaging* 2022;**15**:e013661. doi:10.1161/CIRCIMAGING.121.013661
5. Boscolo Berto M, Spano G, Wagner B, Bernhard B, Häner J, Huber AT, *et al*. Takotsubo cardiomyopathy after mRNA COVID-19 vaccination. *Heart Lung Circ* 2021;**30**:e119-e120. doi:10.1016/j.hlc.2021.06.521
6. Fearon C, Parwani P, Gow-Lee B, Abramov D. Takotsubo syndrome after receiving the COVID-19 vaccine. *J Cardiol Cases* 2021;**24**:223-226. doi:10.1016/j.jccase.2021.08.012
7. Crane P, Wong C, Mehta N, Barlis P. Takotsubo (stress) cardiomyopathy after ChAdOx1 nCoV-19 vaccination. *BMJ Case Rep* 2021;**14**:e246580. doi:10.1136/bcr-2021-246580
8. Vidula MK, Ambrose M, Glassberg H, Chokshi N, Chen T, Ferrari VA, *et al*. Myocarditis and other cardiovascular complications of the mRNA-based COVID-19 vaccines. *Cureus* 2021;**13**:e15576. doi:10.7759/cureus.15576
9. Beshai R, Lee JJ. Unusual case of takotsubo cardiomyopathy secondary to COVID-19 vaccine: Case report and literature review. *Cureus* 2022;**14**:e25398. doi:10.7759/cureus.25398
10. Gill JR, Tashjian R, Duncanson E. Autopsy histopathologic cardiac findings in 2 adolescents following the second COVID-19 vaccine dose. *Arch Pathol Lab Med* 2022;**146**:925-929. doi:10.5858/arpa.2021-0435-SA
11. Jani C, Leavitt J, Al Omari O, *et al*. COVID-19 vaccine-associated takotsubo cardiomyopathy. *Am J Ther* 2021;**28**:361-364. doi:10.1097/MJT.0000000000001379
12. Santoro F, Ieva R, Ferraretti A, Carpagnano G, Pappalardo I, de Gennaro L, *et al*. Tako-Tsubo cardiomyopathy after influenza vaccination. *Int J Cardiol* 2013;**167**:e51-e52. doi:10.1016/j.ijcard.2013.03.147
13. Singh K, Marinelli T, Horowitz JD. Takotsubo cardiomyopathy after anti-influenza vaccination: Catecholaminergic effects of immune system. *Am J Emerg Med* 2013;**31**:1627.e1621-1627.e1624. doi:10.1016/j.ajem.2013.06.039
14. Tedeschi A, Camilli M, Ianni U, Tavecchia G, Palazzini M, Cartella I, *et al*. Takotsubo syndrome after BNT162b2 mRNA Covid-19 vaccine: Emotional or causative relationship with vaccination? *Int J Cardiol Heart Vasc* 2022;**40**:101002. doi:10.1016/j.ijcha.2022.101002
15. Sakamoto T, Kagawa Y, Endo A, Tanabe K. Intense emotional stress over potential coronavirus disease vaccination side effects leads to takotsubo cardiomyopathy. *Circ Res* 2021;**3**:476-477. doi:10.1253/circrep.CR-21-0075
16. Oguri S, Maeda T, Kishimoto S, Kondo H, Ihara K. An adolescent boy with takotsubo cardiomyopathy after COVID-19 vaccination. *Pediatr Int* 2022;**64**:e15321. doi:10.1111/ped.15321
17. Minciullo PL, Amato G, Vita F, Pioggia G, Gangemi S. ATAK complex (adrenaline, takotsubo, anaphylaxis, and Kounis hypersensitivity-associated coronary syndrome) after COVID-19 vaccination and review of the literature. *Vaccine* 2023;**11**:322. doi:10.3390/vaccines11020322
18. Ceresa F, Micari A, Trifirò MP, Costa F, Vizzari G, Giuseppe AV, *et al*. Echocardiographic evolution of posterolateral left ventricular aneurysm with normal coronary arteries in patient recently COVID-19 vaccinated: A complicated atypical case of takotsubo syndrome vaccine-related? *J Cardiovasc Echogr* 2022;**32**:177-179. doi:10.4103/jcecho.jcecho_27_22
19. Chen YC, Lin GM, Takimoto E, *et al*. Takotsubo syndrome occurring after mRNA COVID-19 vaccination in a patient with Graves' disease. *Medicina (Kaunas)* 2022;**59**:94. doi:10.3390/medicina59010094
20. Rahimi O, Varada N, Palma C, al-Taweel O, Lei K, Hawwass D, *et al*. Takotsubo cardiomyopathy in a vaccinated patient with severe COVID-19. *Kansas J Med* 2022;**15**:255-256. doi:10.17161/kjm.vol15.16456
21. Elikowski W, Małek-Elikowska M, Lisiecka M, Trypuć Z, Mozer-Lisewska I. Takotsubo cardiomyopathy triggered by influenza B. *Pol Merkur Lekarski* 2018;**45**:67-70.
22. Tutor A, Unis G, Ruiz B, Bolaji OA, Bob-Manuel T. Spectrum of suspected cardiomyopathy due to COVID-19: A case series. *Curr Probl Cardiol* 2021;**46**:100926. doi:10.1016/j.cpcardiol.2021.100926
23. Tan NYL, Anavekar NS, Wiley BM. Concomitant myopericarditis and takotsubo syndrome following immune checkpoint inhibitor therapy. *BMJ Case Rep* 2020;

13:e235265. doi:10.1136/bcr-2020-235265
24. Lee K, Rahimi O, Gupta N, Ahsan C. Complete AV block in vaccinated COVID-19 patient. *Case Rep Cardiol* 2022;**2022**:9371818. doi:10.1155/2022/9371818
25. Chouchana L, Blet A, Al-Khalaf M, *et al*. Features of inflammatory heart reactions following mRNA COVID-19 vaccination at a global level. *Clin Pharmacol Ther* 2022;**111**:605-613. doi:10.1002/cpt.2499
26. Fazlollahi A, Zahmatyar M, Noori M, Nejadghaderi SA, Sullman MJM, Shekarriz-Foumani R, *et al*. Cardiac complications following mRNA COVID-19 vaccines: A systematic review of case reports and case series. *Rev Med Virol* 2022;**32**:e2318. doi:10.1002/rmv.2318
27. Ghadri JR, Wittstein IS, Prasad A, Sharkey S, Dote K, Akashi YJ, *et al*. International expert consensus document on takotsubo syndrome (part II): Diagnostic workup, outcome, and management. *Eur Heart J* 2018;**39**:2047-2062. doi:10.1093/eurheartj/ehy077
28. Kounis NG, Mplani V, Koniari I. Autopsy histopathologic cardiac findings in 2 adolescents following the second COVID-19 vaccine dose: Cytokine storm, hypersensitivity, or something else. *Arch Pathol Lab Med* 2022;**146**:924. doi:10.5858/arpa.2022-0102-LE
29. Ghadri JR, Wittstein IS, Prasad A, Sharkey S, Dote K, Akashi YJ, *et al*. International expert consensus document on takotsubo syndrome (part I): Clinical characteristics, diagnostic criteria, and pathophysiology. *Eur Heart J* 2018;**39**:2032-2046. doi:10.1093/eurheartj/ehy076
30. Lyon AR, Bossone E, Schneider B, Sechtem U, Citro R, Underwood SR, *et al*. Current state of knowledge on takotsubo syndrome: A position statement from the Taskforce on Takotsubo Syndrome of the Heart Failure Association of the European Society of Cardiology. *Eur J Heart Fail* 2016;**18**:8-27. doi:10.1002/ejhf.424
31. Lazaros G, Klein AL, Hatziantoniou S, Tsioufis C, Tsakris A, Anastassopoulou C. The novel platform of mRNA COVID-19 vaccines and myocarditis: Clues into the potential underlying mechanism. *Vaccine* 2021;**39**:4925-4927. doi:10.1016/j.vaccine.2021.07.016
32. Arunachalam PS, Scott MKD, Hagan T, Li C, Feng Y, Wimmers F, *et al*. Systems vaccinology of the BNT162b2 mRNA vaccine in humans. *Nature* 2021;**596**:410-416. doi:10.1038/s41586-021-03791-x
33. Pardi N, Hogan MJ, Naradikian MS, Parkhouse K, Cain DW, Jones L, *et al*. Nucleoside-modified mRNA vaccines induce potent T follicular helper and germinal center B cell responses. *J Exp Med* 2018;**215**:1571-1588. doi:10.1084/jem.20171450
34. Sahin U, Karikó K, Türeci Ö. mRNA-based therapeutics—Developing a new class of drugs. *Nat Rev Drug Discov* 2014;**13**:759-780. doi:10.1038/nrd4278
35. Astier A, Barton Pai A, Bissig M, Crommelin DJA, Flühmann B, Hecq JD, *et al*. How to select a nanosimilar. *Ann N Y Acad Sci* 2017;**1407**:50-62. doi:10.1111/nyas.13382
36. Li C, Lee A, Grigoryan L, Arunachalam PS, Scott MKD, Trisal M, *et al*. Mechanisms of innate and adaptive immunity to the Pfizer-BioNTech BNT162b2 vaccine. *Nat Immunol* 2022;**23**:543-555. doi:10.1038/s41590-022-01163-9
37. Farsakoglu Y, Palomino-Segura M, Latino I, Zanaga S, Chatziandreou N, Pizzagalli DU, *et al*. Influenza vaccination induces NK-cell-mediated type-II IFN response that regulates humoral immunity in an IL-6-dependent manner. *Cell Rep* 2019;**26**:2307-2315.e2305. doi:10.1016/j.celrep.2019.01.104
38. Münzel T, Templin C, Cammann VL, Hahad O. Takotsubo syndrome: Impact of endothelial dysfunction and oxidative stress. *Free Radic Biol Med* 2021;**169**:216-223. doi:10.1016/j.freeradbiomed.2021.03.033
39. Templin C, Ghadri JR, Diekmann J, Napp LC, Bataiosu DR, Jaguszewski M, *et al*. Clinical features and outcomes of takotsubo (stress) cardiomyopathy. *N Engl J Med* 2015;**373**:929-938. doi:10.1056/NEJMoa1406761
40. Vallabhajosyula S, Barsness GW, Herrmann J, Anavekar NS, Gulati R, Prasad A. Comparison of complications and in-hospital mortality in takotsubo (apical ballooning/stress) cardiomyopathy versus acute myocardial infarction. *Am J Cardiol* 2020;**132**:29-35. doi:10.1016/j.amjcard.2020.07.015
41. José SG, Carrizales-Sepúlveda EF, Vera-Pineda R, Morales-Rendón EJ, de Jesús Ortiz-Corona J, Flores-Ramírez R. Takotsubo cardiomyopathy and COVID-19: A case report and literature review. *Curr Cardiol Rev* 2023;**19**:92-100. doi:10.2174/1573403X18666220818155039
42. Techasatian W, Nishimura Y, Nagamine T, Ha G, Huang R, Shah P, *et al*. Characteristics of takotsubo cardiomyopathy in patients with COVID-19: Systematic scoping review. *Am Heart J* 2022;**13**:100092. doi:10.1016/j.ahjo.2022.100092
43. Dweck MR, Bularga A, Hahn RT, Bing R, Lee KK, Chapman AR, *et al*. Global evaluation of echocardiography in patients with COVID-19. *Eur Heart J Cardiovasc Imaging* 2020;**21**:949-958. doi:10.1093/ehjci/jeaa178
44. Yugar-Toledo JC, Yugar LBT, Sedenho-Prado LG, Schreiber R, Moreno H. Pathophysiological effects of SARS-CoV-2 infection on the cardiovascular system and its clinical manifestations—A mini review. *Front Cardiovasc Med* 2023;**10**:1162837. doi:10.3389/fcvm.2023.1162837